

EE2213 Introduction to Artificial Intelligence

Revision 2

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Q1. True or false: In a linear program with two variables, if the slope of the level set of the objective function is identical to the slope of one of the constraints, we will always have multiple optimal solutions.

- a) True
- b) False

Recap: LP Problem formulation

x_1, x_2 : Decision variables

Maximize $150x_1 + 200x_2$ ← Objective function

subject to:

$$\begin{aligned}x_1 + x_2 &\leq 200 \\9x_1 + 3x_2 &\leq 1240 \\12x_1 + 16x_2 &\leq 2660 \\x_1, x_2 &\geq 0\end{aligned}$$

← Constraints

In linear program: **objective function** + **constraints** are **all linear**

Typically (not always): **variables are non-negative**

Linear programming uses **only non-strict inequalities** ($\leq, \geq$) and equalities ($=$), **not strict inequalities** ($<, >$)

If variables are integer: system called Integer Linear Programme (ILP)*.

* ILP involves more advanced mathematical complexity, so we won't cover it in lectures, but you'll gain hands-on experience with small-scale integer programs during tutorials.

Recap: Linear programs (LP)

An optimization problem that aims to **maximize or minimize a linear objective function**, subject to **linear equality and/or inequality constraints**.

$$\begin{array}{ll}\text{Maximize} & \mathbf{c}^T \mathbf{x} \\ \text{Subject to} & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq 0\end{array}$$

Solving LPs:

Graphical method: applicable when there are only two variables.

Simplex algorithm: efficient for large-scale LPs

Fundamental theorem: Optimal solutions occur on the boundary of the feasible region — typically at its vertices (or along an edge between vertices).

Quick question

Select all options that are **linear programs**.

(A) Minimize $3x + 4y$

$$\begin{aligned} \text{s.t.} \quad & x + 2y \geq 8 \\ & 5x + y = 10 \\ & x, y \geq 0 \end{aligned}$$

(B) Maximize $7u + 9v$

$$\begin{aligned} \text{s.t.} \quad & 2u - v \leq 6 \\ & uv \leq 10 \\ & u, v \geq 0 \end{aligned}$$

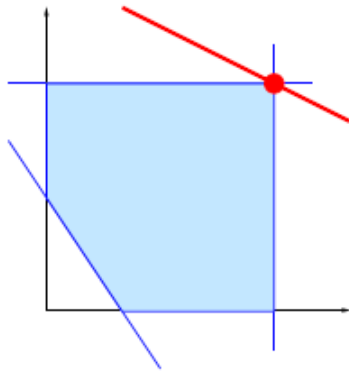
(C) Maximize $5p + q$

$$\begin{aligned} \text{s.t.} \quad & 3p + 2q < 12 \\ & p, q \geq 0 \end{aligned}$$

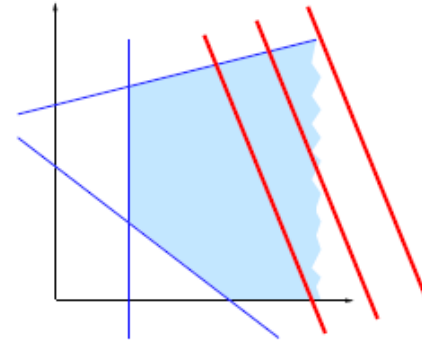
(D) Minimize $4a - b$

$$\begin{aligned} \text{s.t.} \quad & -a + 3b \geq 9 \\ & 2a + b \leq 7 \\ & a, b \geq 0 \end{aligned}$$

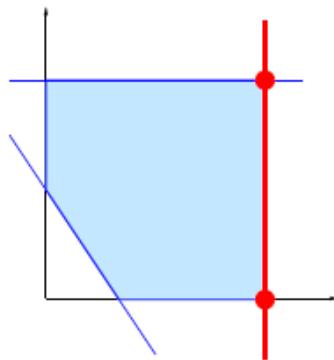
Recap: Different cases of LP outcomes



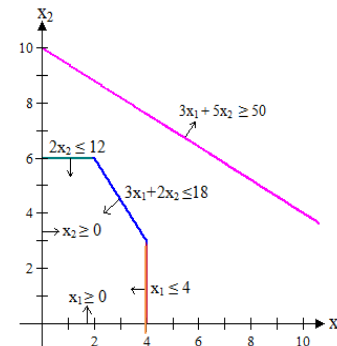
(a) Single optimal solution



(c) No optimal solution



(b) Infinite number of optimal solution



(d) No optimal solution

Example: Multiple optimal solution

An LP may have multiple optimal solutions.

Minimize $z = x_1 + 2x_2$

subject to:

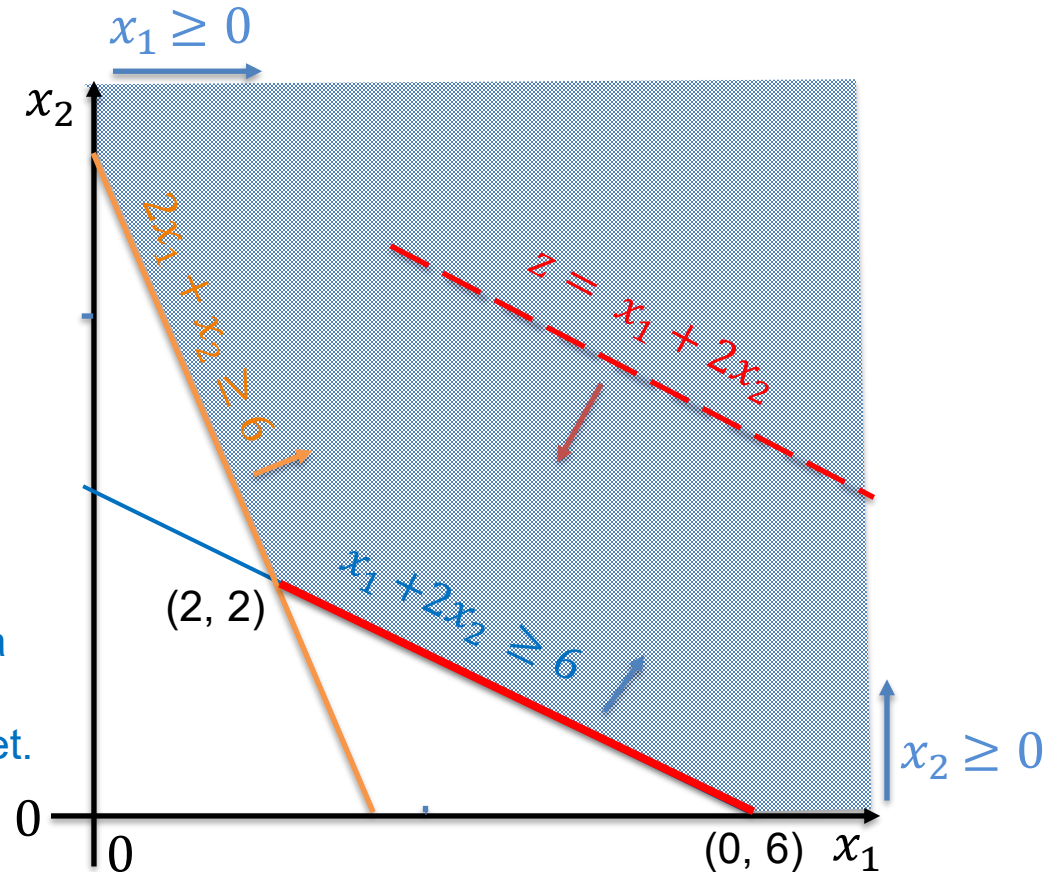
$$x_1 + 2x_2 \geq 6$$

$$2x_1 + x_2 \geq 6$$

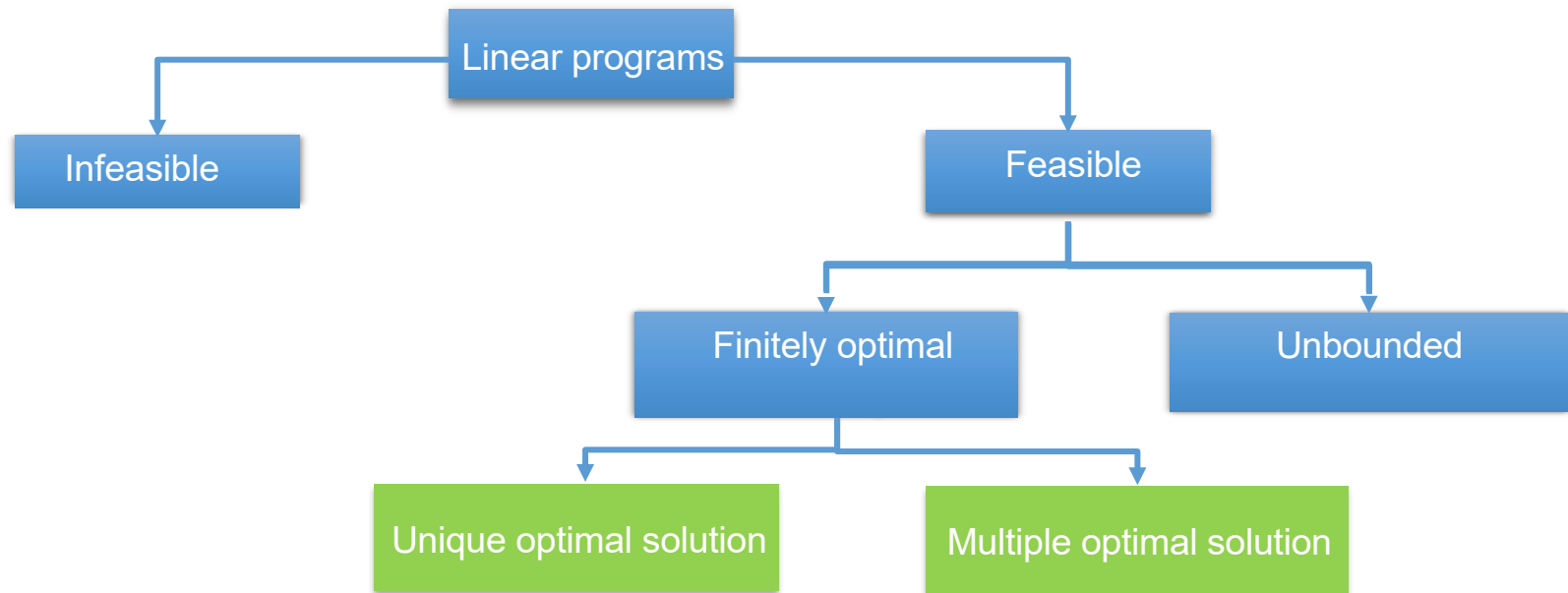
$$x_1, x_2 \geq 0$$

$$z_{\min} = 6$$

Finitely Optimal: Objective reaches a finite maximum/minimum, even if optimal solutions form a continuous set.



Recap: Different cases of LP & outcomes



Recap: Three types of LPs

Every LP falls into one of these cases:

Infeasible — No solution satisfies all constraints.

Unbounded — Objective can improve indefinitely within the feasible region.

Finitely optimal — An optimal solution exists and is finite.

If finitely optimal, the solution can be:

Unique, or

Multiple optimal solutions along an edge or face of the feasible region.

Q1. True or false: In a linear program with two variables, if the slope of the level set of the objective function is identical to the slope of one of the constraints, we will always have multiple optimal solutions.

- a) True
- b) False ✓

Q2 Case 1: Optimal edge coincidence (leads to multiple solutions)

Minimize $z = x_1 + 2x_2$

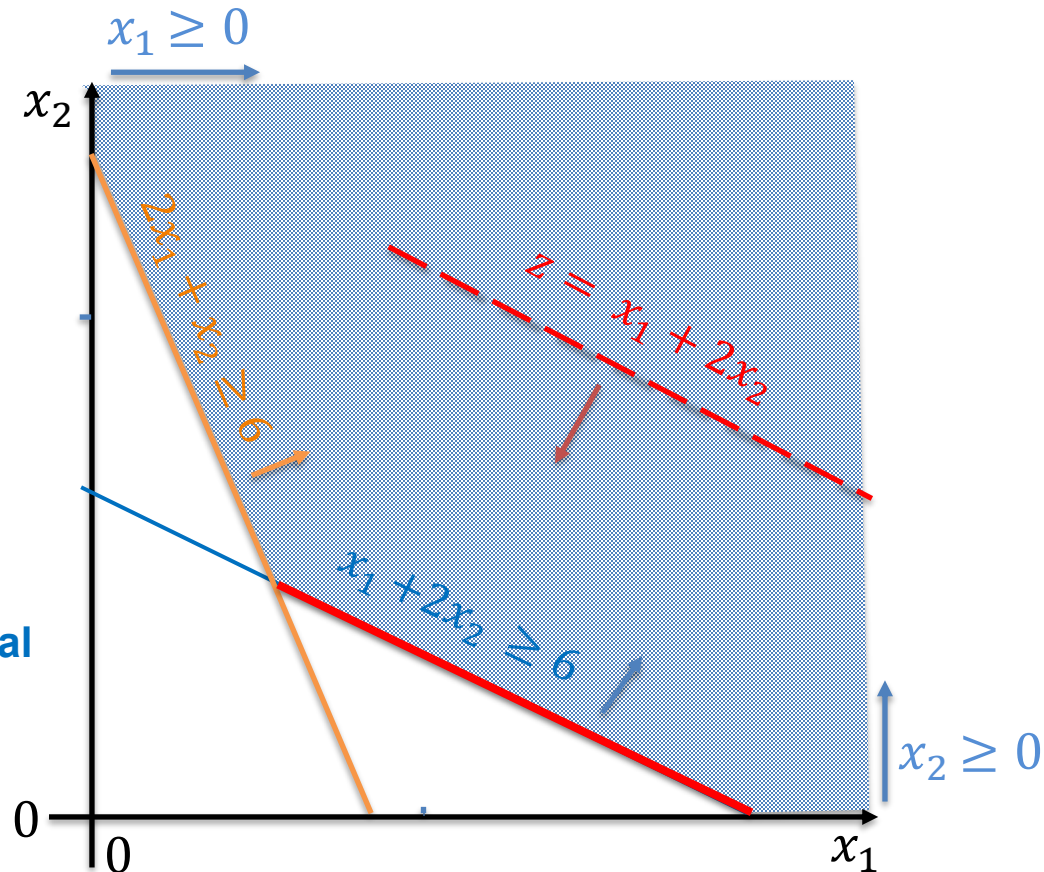
subject to:

$$x_1 + 2x_2 \geq 6$$

$$2x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

Finitely optimal with multiple optimal solution: Objective reaches a finite maximum/minimum, even if optimal solutions form a continuous set.



Q2 Case 2: Parallel but non-active constraint (unique solution)

Maximize $z = x_1 + x_2$

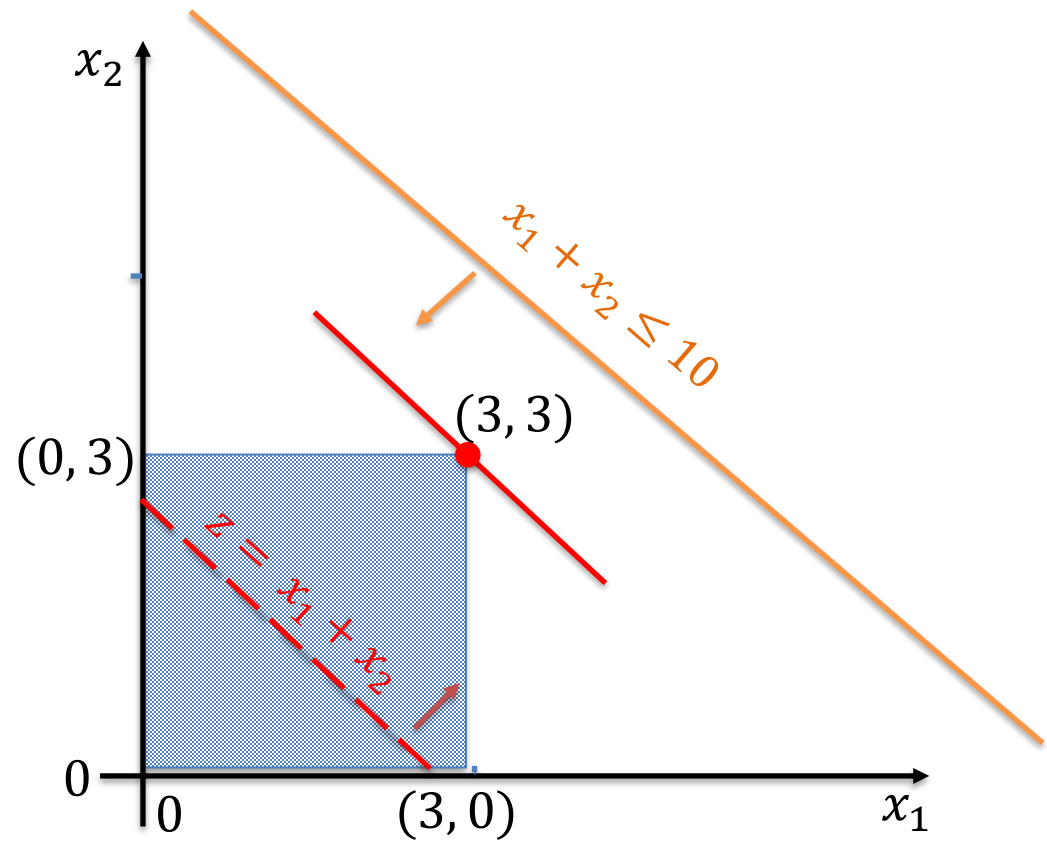
subject to:

$$x_1 + x_2 \leq 10$$

$$x_1, x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

**Finitely optimal with unique
optimal solution**



Q2 Case 3: Parallel but optimum is a single vertex (unique solution)

Maximize $z = x_1 + x_2$

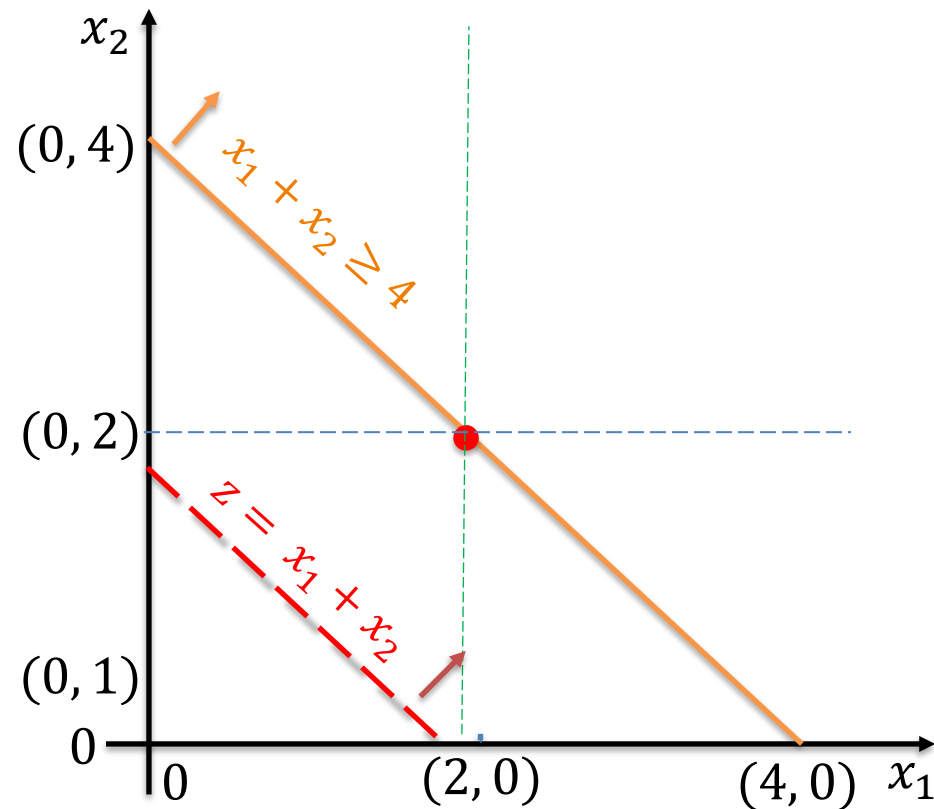
subject to:

$$x_1 + x_2 \geq 4$$

$$x_1 \leq 2$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



Quick question

True or False: In a linear program with two variables, if the slope of the level set of the objective function is different from all constraint boundary slopes, the optimal solution must occur at a single vertex of the feasible region.

- a) True
- b) False

Q2. Consider the following linear program (LP), select all the options that are correct. (More than one answer may be correct.)

Maximize: $z = 3x_1 + 4x_2$

Subject to:

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

- a) The maximum value of the objective function is 16.
- b) If we remove the constraint $x_1, x_2 \geq 0$, it is still an LP problem.
- c) If we remove the constraint $x_1 + x_2 \leq 4$, it is still a feasible LP problem.
- d) If the constraints were changed to: $x_1 + x_2 < 4$, it is still a linear program.
- e) If we changed the problem's objective function to: Minimize $z = -3x_1 - 4x_2$, then the optimal solution for (x_1, x_2) remains unchanged.
- f) $(-1, 4)$ is a feasible but not the optimal solution for the above LP problem.

Q2. Consider the following linear program (LP), select all the options that are correct. (More than one answer may be correct.)

Maximize: $z = 3x_1 + 4x_2$

Subject to:

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

- a) The maximum value of the objective function is 16. ✓
- b) If we remove the constraint $x_1, x_2 \geq 0$, it is still an LP problem. ✓
- c) If we remove the constraint $x_1 + x_2 \leq 4$, it is still a feasible LP problem. ✓
- d) If the constraints were changed to: $x_1 + x_2 < 4$, it is still a linear program.
- e) If we changed the problem's objective function to: Minimize $z = -3x_1 - 4x_2$, then the optimal solution for (x_1, x_2) remains unchanged. ✓
- f) $(-1, 4)$ is a feasible but not the optimal solution for the above LP problem.

Q3. Tournament Transportation Optimization



Your student club is organizing a massive inter-university gaming tournament with 400 participants. You need to arrange transportation from campus to the esports arena using two types of awesome vehicles:

Party Bus:

Capacity: 50 players, Cost: \$800 per bus, Requires 1 driver, Max 10 available
Comes with RGB lighting, snack stations, and a built-in sound system.

Gamer Van:

Capacity: 40 players, Cost: \$600 per van, Requires 1 driver, Max 8 available (they're popular!)

Equipped with individual charging ports, headphone sanitizers, and premium seating.

Constraints you face:

Driver Shortage: You will use your own drivers from your club. Only 9 club members have driver licenses, and there is no extra budget to hire external drivers.

Budget Cap: The total rental cost cannot exceed \$6,000 (we know how club funds work!).

Non-Negotiable: You must transport all 400 participants.

Vehicle Mix Policy: The club president decrees that at least 40% of the vehicles must be Gamer Vans (to ensure enough charging ports).

Your mission is to minimize the total rental cost so you can afford the best prizes for the winners.

Let decision variables:

B = number of Party Buses used; **V** = number of Gamer Vans used

Q3. Tournament Transportation Optimization

$$\text{Min } 800B + 600V$$

s.t.:

$$B + V \leq 9$$

$$50B + 40V \geq 400$$

$$B \leq 10$$

$$V \leq 8$$

$$800B + 600V \leq 6000$$

$$3V \geq 2B$$

$$B, V \geq 0$$

$$B, V \in \mathbb{Z}$$

Q4. The given linear program is:

Maximize $z = 3x_1 + 2x_2$
subject to:

$$x_1 + x_2 \leq 5$$

$$2x_1 + x_2 \geq 12$$

$$-x_1 + 2x_2 \leq 4$$

$$x_1 - x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

- (a) Bounded and feasible
- (b) Feasible but not bounded
- (c) Bounded but not feasible
- (d) Neither bounded nor feasible

```
# Import libraries
import cvxpy as cp # For solving linear programs

# 1. Define variables
x_1 = cp.Variable() # defines a continuous decision variable x_1.
x_2 = cp.Variable() # defines a continuous decision variable x_2.

# 2. Define objective
objective = cp.Maximize(3 * x_1 + 2 * x_2)

# 3. Define constraints
constraints = [
    x_1 >= 0,
    x_2 >= 0,
    x_1 + x_2 <= 5,
    2 * x_1 + x_2 >= 12,
    -x_1 + 2 * x_2 <= 4,
    x_1 - x_2 <= 3
]

# 4. Create a CVXPY problem object
prob = cp.Problem(objective, constraints)

# 5. Solve the LP problem
prob.solve()

# 6. Results
x1_opt = x_1.value # Get the optimal value of decision variable x_1
x2_opt = x_2.value # Get the optimal value of decision variable x_2
min_obj = prob.value # Get the minimum from the objective function
problem_status = prob.status # Get the solution status

# Output results
print("\nSolution Status:", problem_status)
print("x1:", x1_opt)
print("x2:", x2_opt)
print("Minimum objective value:", min_obj)
```

```
Solution Status: infeasible
x1: None
x2: None
Minimum objective value: -inf
```

Q4. The given linear program is:

Maximize $z = 3x_1 + 2x_2$
subject to:

$$x_1 + x_2 \leq 5$$

$$2x_1 + x_2 \geq 12$$

$$-x_1 + 2x_2 \leq 4$$

$$x_1 - x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

- (a) Bounded and feasible
- (b) Feasible but not bounded
- (c) Bounded but not feasible
- (d) Neither bounded nor feasible ✓

Q5. Which of the following functions are convex?
Select all that apply.

a) $f(x) = x^2 - 4x + 10, x \in \mathcal{R}$

b) $f(x) = |x - 3|, x \in \mathcal{R}$

c) $f(x) = -x^2, x \in \mathcal{R}$

d) $f(x) = \sqrt{x}, x \in (0, +\infty)$

Recap: Convex function

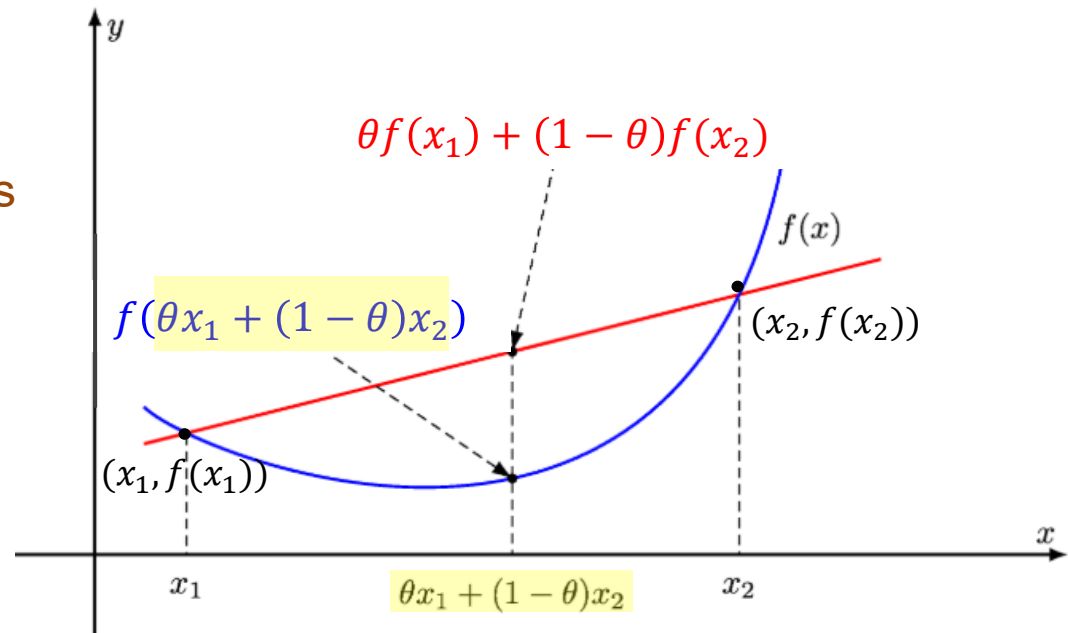
Let C be a convex set. A function $f: C \rightarrow \mathbb{R}$ is convex in C if $\forall x_1, x_2 \in C, \forall \theta \in [0,1]$,

$$f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2)$$

If $C = \mathbb{R}^n$, we simply say f is a convex function.

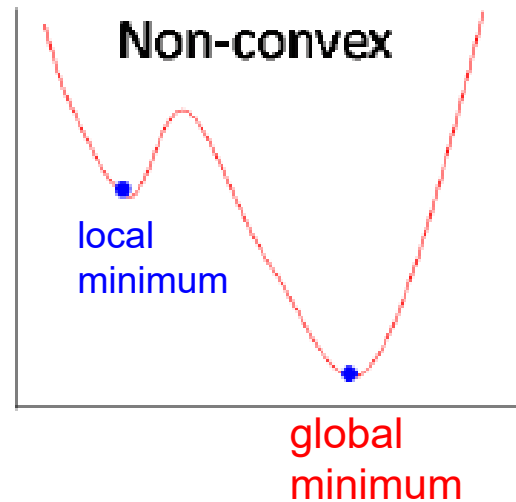
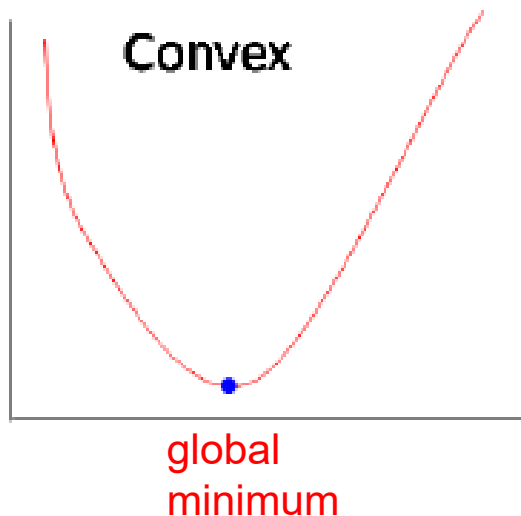
Any chord between two points on the function always lies above the function.

Value in the middle point is lower than the average of the two endpoint values.

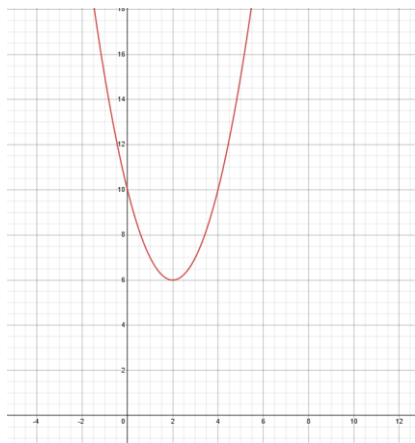


Recap: Convex function properties

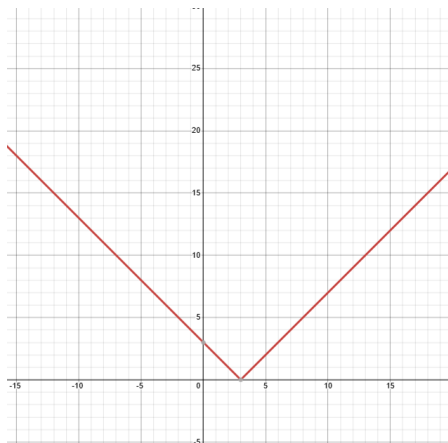
- Sum of convex functions is convex
 - $af(x) + bg(x)$ is convex for convex functions f, g and $a, b > 0$.
- Convexity is preserved under affine transformation
 - If $f(\mathbf{x}) = g(\mathbf{Ax} + \mathbf{b})$, g is convex, then $f(\mathbf{x})$ is convex.
where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$
- Any **local minimum** is a **global minimum**



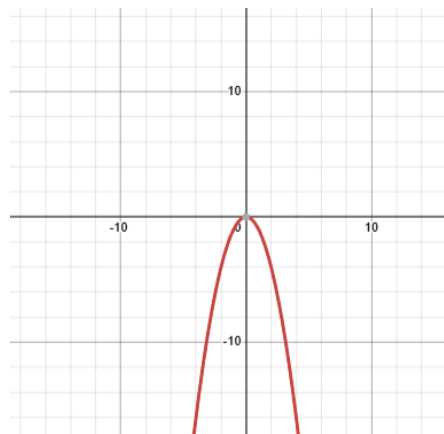
Q5 Method 1: Graphing



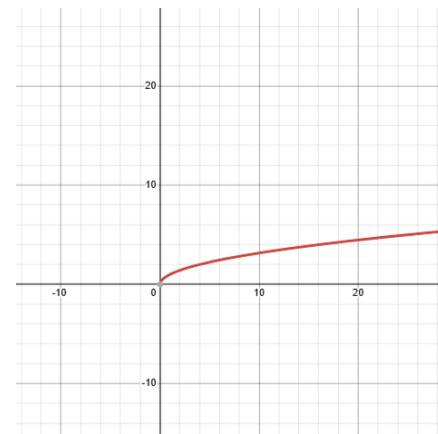
$$f(x) = x^2 - 4x + 10$$



$$f(x) = |x - 3|$$



$$f(x) = -x^2$$



$$f(x) = \sqrt{x}$$

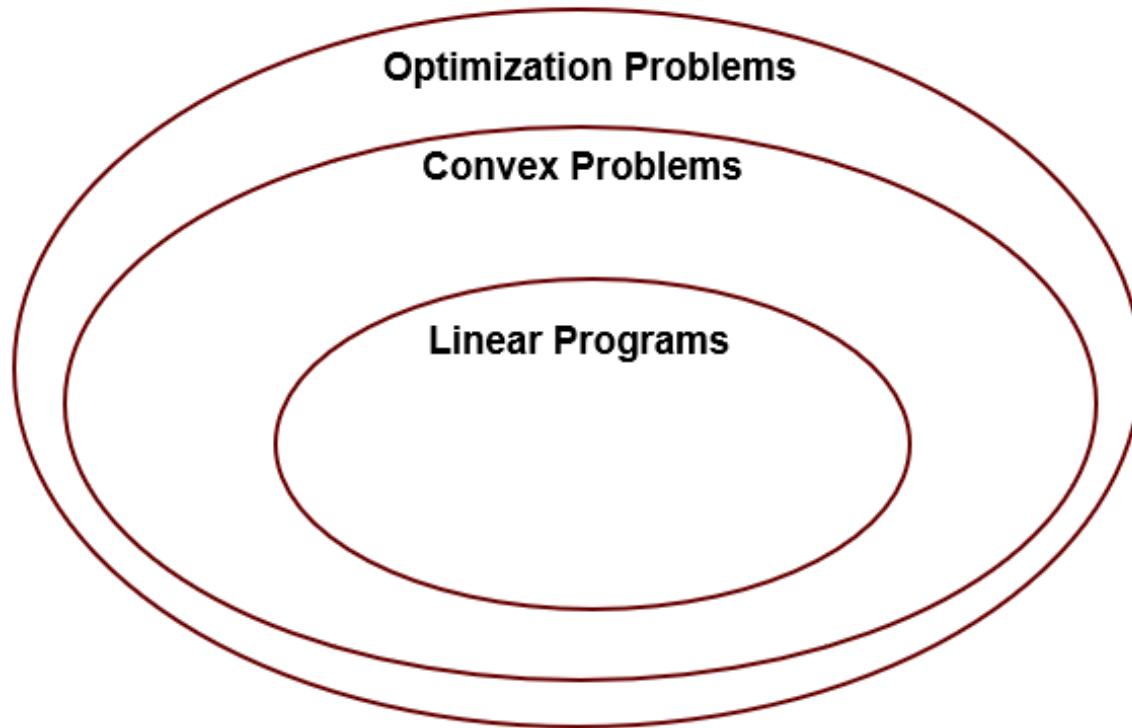
Q5 Method 2: The second derivative test (for single-variable functions)

If the second derivative, $f''(x)$ is **always greater than or equal to 0** ($f''(x) \geq 0$) *throughout the function's domain*, then the function is **convex**, if $f''(x) \leq 0$, then the function is concave.

Example 1: For the function $f(x) = (x - 2)^2$,
 $f''(x) = 2$, which is always ≥ 0 , so the function is convex.

Example 2: For the function $f(x) = \log(x)$, $x \in (0, +\infty)$
 $f'(x) = \frac{1}{x}$, $f''(x) = -\frac{1}{x^2}$, which is always < 0 , so the function is concave.

Linear programs vs convex problems



Q5. Which of the following functions are convex?
Select all that apply.

a) $f(x) = x^2 - 4x + 10, x \in \mathcal{R}$ ✓

b) $f(x) = |x - 3|, x \in \mathcal{R}$ ✓

c) $f(x) = -x^2, x \in \mathcal{R}$

d) $f(x) = \sqrt{x}, x \in (0, +\infty)$

Q6. Which of the following statements about gradient descent are correct? Select all that apply.

- a) Gradient descent updates the function's variable by moving in the opposite direction of the gradient — that is, in the direction of steepest decrease of the function.
- b) For an unconstrained convex optimization problem with a differentiable objective function, gradient descent will converge to the global minimum if the learning rate is chosen appropriately.
- c) If the learning rate is too large, gradient descent might diverge or oscillate instead of converging.
- d) For non-convex functions, gradient descent may get stuck in local minima, but under certain conditions it can find reasonably good solutions.
- e) Gradient descent can work for any differentiable function, not just convex ones.

Recap: Gradient descent algorithm

Gradient Descent:

Initialize $\mathbf{x}_0$ and learning rate η

while true **do**

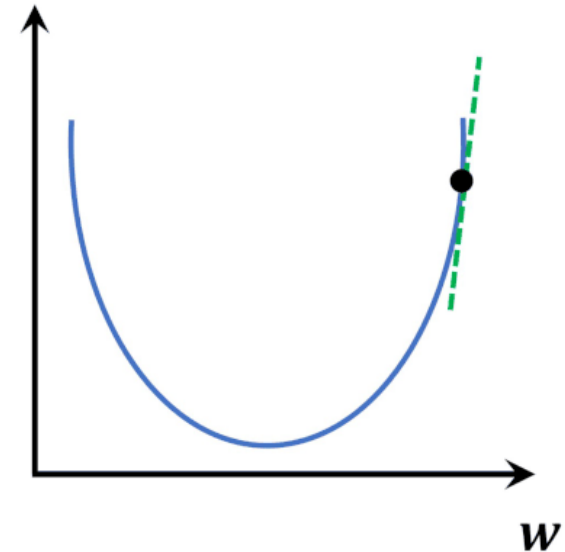
 Compute $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k - \eta \nabla_{\mathbf{x}} f(\mathbf{x}_k)$

if converge **then**

return $\mathbf{x}_{k+1}$

end

end



- Gradient descent is a widely used iterative optimization algorithm that, when applied to **convex functions**, is guaranteed to converge to the **global minimum**.
- It solves convex optimization problems by consistently moving in the direction of the **steepest decrease**.

Animation sources: <https://datahacker.rs/003-pytorch-how-to-implement-linear-regression-in-pytorch/>

Convex optimization: How to solve

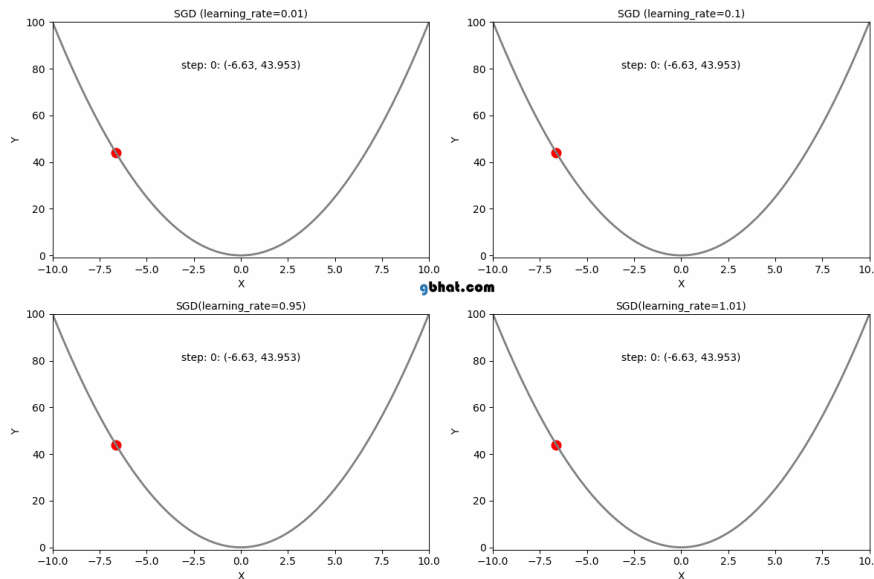
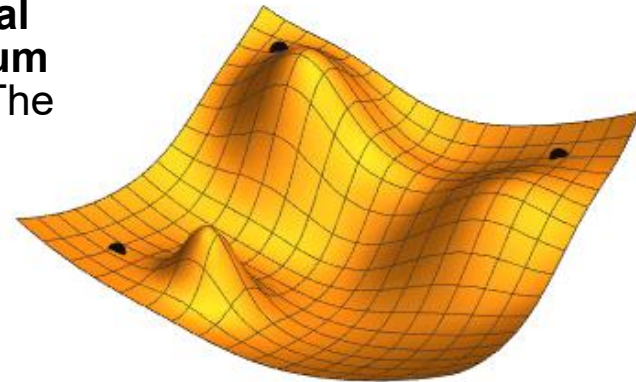
- **Unconstrained** and **differentiable** convex functions:
Gradient descent converges to the **global minimum** with a sufficiently small step size (i. e., learning rate η).
- **Constrained** and **differentiable** convex functions:
Projected gradient descent is used when the domain is constrained to a convex set. (*not required in exam*)
- **Non-differentiable** convex functions (*not covered in this course*):
 ϵ -subgradient methods can be applied

* In non-convex functions, gradient descent might get stuck in local minima but sometimes finds good solutions in practice.

Limitation of Gradient Descent

• **Gets stuck in local minima:** For *non-convex functions*, gradient descent can get stuck in a **local minimum**, and may fail to find the **global minimum** (the lowest possible point of the entire function). The starting point for the algorithm can significantly influence which minimum it converges to.

• **Initialization & learning rate sensitivity:** The outcome depends heavily on the starting point (initialization) and the learning rate (step size). A poor choice of either can lead to suboptimal results.



Source: https://commons.wikimedia.org/wiki/File:Gradient_descent.gif
https://gbhat.com/machine_learning/gradient_descent_learning_rates.html

Q6. Which of the following statements about gradient descent are correct? Select all that apply.

- a) Gradient descent updates the function's variable by moving in the opposite direction of the gradient — that is, in the direction of steepest decrease of the function. ✓
- b) For an unconstrained convex optimization problem with a differentiable objective function, gradient descent will converge to the global minimum if the learning rate is chosen appropriately. ✓
- c) If the learning rate is too large, gradient descent might diverge or oscillate instead of converging. ✓
- d) For non-convex functions, gradient descent may get stuck in local minima, but under certain conditions it can find reasonably good solutions. ✓
- e) Gradient descent can work for any differentiable function, not just convex ones. ✓

Q7.

The given linear program is:

Maximize $z = 7x_1 + 2x_2 + 6x_3$
subject to:

$$2x_1 + x_2 + 3x_3 \leq 14$$

$$-x_1 + 4x_2 + x_3 \geq 3$$

$$3x_1 - 2x_2 + 2x_3 \leq 12$$

$$x_1 + 5x_2 + 2x_3 \leq 20$$

$$x_1 + x_2 + 3x_3 \geq 5$$

$$x_1, x_2, x_3 \geq 0$$

Solve this linear program. Provide the maximum objective value. Round your answer to 2 decimal places.

```
# Import libraries
import cvxpy as cp # For solving linear programs

# 1. Define variables
x_1 = cp.Variable() # defines a continuous decision variable x_1.
x_2 = cp.Variable() # defines a continuous decision variable x_2.
x_3 = cp.Variable() # defines a continuous decision variable x_23.

# 2. Define objective
objective = cp.Maximize(7 * x_1 + 2 * x_2 + 6 * x_3 )

# 3. Define constraints
constraints = [
    x_1 >= 0,
    x_2 >= 0,
    x_3 >= 0,
    2 * x_1 + x_2 + 3 * x_3 <= 14,
    -x_1 + 4 * x_2 + x_3 >= 3,
    3 * x_1 - 2 * x_2 + 2 * x_3 <= 12,
    x_1 + 5 * x_2 + 2 * x_3 <= 20,
    x_1 + x_2 + 3 * x_3 >= 5
]

# 4. Create a CVXPY problem object
prob = cp.Problem(objective, constraints)

# 5. Solve the LP problem
prob.solve()

# 6. Results
x1_opt = x_1.value # Get the optimal value of decision variable x_1
x2_opt = x_2.value # Get the optimal value of decision variable x_2
x3_opt = x_3.value # Get the optimal value of decision variable x_2
min_obj = prob.value # Get the minimum from the objective function
problem_status = prob.status # Get the solution status

# Output results
print("\nSolution Status:", problem_status)
print("x1:", x1_opt)
print("x2:", x2_opt)
print("x3:", x3_opt)
print("Minimum objective value:", min_obj)
```

```
x1: 5.714285713572853
x2: 2.5714285717450025
x3: 2.4148555082345343e-10
Minimum objective value: 45.14285713994889
```

Q7. Method 2 (using PuLP)

```
# Import PuLP library
from pulp import LpMaximize, LpProblem, LpVariable, LpStatus

# 1. Define the problem
prob = LpProblem("Maximize_Objective", LpMaximize)

# 2. Define decision variables
x1 = LpVariable("x1", lowBound=0) # x1 >= 0
x2 = LpVariable("x2", lowBound=0) # x2 >= 0
x3 = LpVariable("x3", lowBound=0) # x3 >= 0

# 3. Define objective function
prob += 7 * x1 + 2 * x2 + 6 * x3, "Objective"

# 4. Define constraints
prob += 2 * x1 + x2 + 3 * x3 <= 14
prob += -x1 + 4 * x2 + x3 >= 3
prob += 3 * x1 - 2 * x2 + 2 * x3 <= 12
prob += x1 + 5 * x2 + 2 * x3 <= 20
prob += x1 + x2 + 3 * x3 >= 5

# 5. Solve the problem
prob.solve()

# 6. Results
print("\nSolution Status:", LpStatus[prob.status])
print("x1 =", x1.varValue)
print("x2 =", x2.varValue)
print("x3 =", x3.varValue)
print("Maximum objective value =", prob.objective.value())
```

```
Solution Status: Optimal
x1 = 5.7142857
x2 = 2.5714286
x3 = 0.0
Maximum objective value = 45.1428571
```



All the best for your exams!

Provide your feedback now >>

<https://blue.nus.edu.sg/Blue/>

